Hierarchy of Pseudo-Random Array Codes

Yeow Meng Chee
Dept. Indus. Systems Eng. and Managment
National Univ. of Singapore
Singapore
ymchee@nus.edu.sg

Tuvi Etzion
Dept. of Computer Science
Technion-Israel Institute of Technology
Haifa 320003, Israel
etzion@cs.technion.ac.il

Huimin Lao
Dept. Electrical and Computer Eng.
National Univ. of Singapore
Singapore
huiminl@nus.edu.sg

2025 IEEE International Symposium on Information Theory (ISIT) | 979-8-3315-4399-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISIT63088.2025.11195674

***Abstract*—Pseudo-random arrays are the two-dimensional analog of M-sequences. Pseudo-random array codes are the two-dimensional analog of sequences generated by a product of irreducible polynomials with the same exponent. The union of the arrays in such a code has the window property and the shift-and-add property, implying that these codes are linear. The folding technique is the most basic one for forming such arrays and codes. A new criterion for generating pseudo-random arrays based on folding is given. This new criterion yields pseudo-random arrays with new parameters. A general construction for such array codes is given. It appears that the arrays generated in this construction can be constructed by folding the nonzero sequences generated by a product of irreducible polynomials of the same degree and the same exponent. Two hierarchies of the pseudo-random array codes are provided. In one hierarchy codewords of one code with smaller windows are contained in codewords of another code which stands above him in the hierarchy. The second hierarchy is a partition of the pseudo-random array codes generated by folding into classes based on the polynomial types which participate in their construction.**

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

Generalizations of one-dimensional sequences and codes to higher dimensions are quite natural and fashionable from both theoretical and practical points of view. Such generalizations were considered for various structures such as error-correcting codes [1], [22], burst-correcting codes [2], [3], [10], [11], constrained codes [25], [27], and de Bruijn sequences [5], [17], [20]. This paper considers a generalization of one-dimensional linear sequences with a window property. For simplicity, only binary arrays and binary sequences are considered in this paper, although some of the results can be generalized to any alphabet whose size is a power of a prime.

Definition 1: A ***de Bruijn array*** (often known as a ***perfect map***) is an $r \times t$ doubly-periodic array (cyclic horizontally and vertically like in a torus), such that each $n \times m$ binary matrix appears exactly once as a window in the array.

Perfect maps were first presented by Reed and Stewart [21] and considered in hundreds of papers which discussed constructions and applications of these arrays. The definition was generalized in [8], [9] for a set of arrays, i.e., a code which contains arrays of the same dimensions, as follows.

Definition 2: A ***de Bruijn array code*** is a set of $r \times t$ doubly-periodic arrays, such that each $n \times m$ matrix appears exactly once as a window in one of the arrays.

The research of Tuvi Etzion was supported in part by the Israeli Science Foundation grant no. 222/19.

Definition 3: A ***shortened de Bruijn array*** (or a ***shortened perfect map***) is an $r \times t$ doubly-periodic array, such that $rt = 2^{nm} - 1$ and each nonzero $n \times m$ matrix appears exactly once as a window in the array. Such an array will be called an $(r, t; n, m)$-SDBA. A ***shortened de Bruijn array code*** $\mathcal{C}$ is a set of $\frac{2^{nm}-1}{rt}$ arrays of size $r \times t$ such that each nonzero $n \times m$ matrix is contained exactly once in one of the arrays. Such a code will be denoted by an $(r, t; n, m)$-SDBAC.

Lemma 1: If $\mathcal{C}$ is an $(r, t; n, m)$-SDBAC of size Δ, then

1. $r > n$ or $r = n = 1$,
2. $t > m$ or $t = m = 1$, and
3. $\Delta rt = 2^{nm} - 1$.

Definition 4: An $(r, t; n, m)$ ***pseudo-random array*** ($(r, t; n, m)$-PRA in short) $\mathcal{A}$ is an $(r, t; n, m)$-SDBA, for which if $\mathcal{A}'$ is a nontrivial shift of $\mathcal{A}$, then $\mathcal{A} + \mathcal{A}'$ is also a nontrivial shift of $\mathcal{A}$, where the addition is performed bit-by-bit. This is the ***shift-and-add property*** of the PRA.

Pseudo-random arrays were constructed first in [16], [18], [26], where properties and applications of such arrays were discussed. They have found various important applications, e.g., they were used in pattern recognition for structured light systems [12], [15], [19], [23], [24]. These applications motivated the current work.

Definition 5: An $(r, t; n, m)$ ***pseudo-random array code*** ($(r, t; n, m)$-PRAC in short) $\mathcal{C}$ is a $(r, t; n, m)$-SDBACs, for which if $\mathcal{A}, \mathcal{B} \in \mathcal{C}$ ($\mathcal{B}$ can also be a nontrivial shift of $\mathcal{A}$), then $\mathcal{A} + \mathcal{B}$ is also a codeword of $\mathcal{C}$. This is the ***shift-and-add property*** of the pseudo-random array codes.

In the current work, constructions for PRAs and PRACs are presented. The codes are formed by folding nonzero sequences generated by a polynomial whose nonzero sequences have the same length. Two novel proof techniques are used to prove the correctness of the suggested constructions. The first one is a generalization to a proof method suggested by MacWilliams and Sloane [16] (Theorem 1). The outcome are pseudo-random arrays with parameters which were not obtained in [16]. The second proof technique is based on the linear span of the bitwise product of sequences generated by two polynomials, each one has nonzero sequences of the same length (Theorem 2).

The rest of this paper is organized as follows. In Section II we present the basic definitions and theory for the sequences and arrays considered here. Section III introduces the main technique in this paper known as folding. PRAs with new parameters are presented in this section. Section IV generalizes the folding technique to be applied on many sequences to form

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 14:54:01 UTC from IEEE Xplore. Restrictions apply.

PRACs. There is also a different method to generate some of these codes which suggests a simpler way to prove the correctness of the construction. In Section V two hierarchies of these codes are presented. For lack of space, most proof are omitted and they will be presented in [4].

II. Preliminaries

Any infinite sequence $\{a_k\}_{k=-n}^{\infty}$ with the initial state $(a_{-n}, a_{-n+1}, \ldots, a_{-1})$, satisfies a linear recursion

$$a_k = \sum_{i=1}^{n} c_i a_{k-i}, \quad c_i \in \{0,1\}, \quad k = 0, 1, \ldots.$$

The ***characteristic polynomial*** of the sequence $\{a_k\}$ is defined by

$$c(x) = 1 - \sum_{i=1}^{n} c_i x^i.$$

We say that the (characteristic) polynomial $c(x)$ ***generates*** the sequence $\{a_k\}$.

The sequences generated by various characteristic polynomials are used in this paper to form the PRACs. Hence, we will define a few properties of these sequences.

We distinguish between acyclic and cyclic sequences. A ***cyclic sequence*** denoted by $[a_0 a_1 \cdots a_{n-1}]$ and an ***acyclic sequence*** (a word) denoted by $(a_0 a_1 \cdots a_{n-1})$. Cyclic sequences can sometimes be viewed as acyclic sequences. For example, when such sequences are concatenated, the sequences are considered as acyclic sequences. The ***length*** of a sequence $\boldsymbol{a} = [a_0, a_1, \cdots, a_{n-1}]$ is n. The length of the sequence $\boldsymbol{a}$ will be denoted by $\ell(\boldsymbol{a})$. **The *period*** of a sequence $\boldsymbol{a} = [a_0, a_1, \cdots, a_{n-1}]$ is the least positive integer π such that $a_i = a_{i+\pi}$ for all $0 \leq i \leq n-1$, where subscripts are taken modulo n. The period of the sequence $\boldsymbol{a}$ will be denoted by $\pi(\boldsymbol{a})$. An integer p such that $a_i = a_{i+p}$ for all $0 \leq i \leq n-1$ is ***a period*** of the sequence. This implies that if a sequence $\boldsymbol{a}$ has length $k\pi$ and $\pi = \pi(\boldsymbol{a})$, then $\kappa\pi$ is a period of $\boldsymbol{a}$ if κ divides k.

Similar definitions are adopted and given for arrays. In particular, the horizontal period of an $r \times t$ array $\mathcal{A}$ is the least integer π such that π is a period of each row of $\mathcal{A}$. Similarly, the vertical period of an $r \times t$ array $\mathcal{A}$ is the least integer π such that π is a period of each column of $\mathcal{A}$. In all the $r \times t$ arrays considered here, the horizontal period will be t and the vertical period will be r.

Let $\boldsymbol{a} = a_0, a_1, \ldots$ and $\boldsymbol{b} = b_0, b_1, \ldots$ be two sequences (either cyclic or acyclic) of the same length. The ***(bitwise) addition*** of $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined as $\boldsymbol{a} + \boldsymbol{b} = (a_0 + b_0, a_1 + b_1, \ldots)$ and The ***bitwise product*** of $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined as $\boldsymbol{a} \cdot \boldsymbol{b} = (a_0 b_0, a_1 b_1, \ldots)$. Furthermore, the bitwise product of two matrices $A = (a_{i,j})$ and $B = (b_{i,j})$ in $\mathbb{F}_2^{r \times t}$ is defined as $A \cdot B = (a_{i,j} b_{i,j})$, where $0 \leq i \leq r-1$ and $0 \leq j \leq t-1$. Similarly, the (bitwise) addition of two matrices is defined.

For two positive integers x and y, let $\gcd(x, y)$ denote the ***greatest common divisor*** of x and y. For two positive integers x and y, let $\ell cm(x, y)$ denotes the ***least common multiplier*** of x and y. When two sequences $\boldsymbol{a}$ and $\boldsymbol{b}$, where $\ell(\boldsymbol{a}) = \pi(\boldsymbol{a})$ and $\ell(\boldsymbol{b}) = \pi(\boldsymbol{b})$ are not of the same period, the bitwise product $\boldsymbol{a} \cdot \boldsymbol{b}$ is defined as the bitwise product of the sequences $(\overbrace{\boldsymbol{a}\boldsymbol{a} \cdots \boldsymbol{a}}^{\mu_1 \text{ copies of } \boldsymbol{a}})$ and $(\overbrace{\boldsymbol{b}\boldsymbol{b} \cdots \boldsymbol{b}}^{\mu_2 \text{ copies of } \boldsymbol{b}})$, where $\mu_1 = \frac{\ell cm(\pi(\boldsymbol{a}), \pi(\boldsymbol{b}))}{\pi(\boldsymbol{a})}$ and $\mu_2 = \frac{\ell cm(\pi(\boldsymbol{a}), \pi(\boldsymbol{b}))}{\pi(\boldsymbol{b})}$. This implies that the sequence $\boldsymbol{a} \cdot \boldsymbol{b}$ has length and period $\ell cm(\pi(\boldsymbol{a}), \pi(\boldsymbol{b}))$. In the same way we define the sequences on which the bitwise addition of $\boldsymbol{a} + \boldsymbol{b}$ is applied.

Proposition 1: If $\boldsymbol{a}$ and $\boldsymbol{b}$ are two sequences generated by the polynomial $f(x)$, then the sequence $\boldsymbol{a} + \boldsymbol{b}$ is also generated by the polynomial $f(x)$.

Proof: The claim follows immediately from the fact that if $\boldsymbol{a}$ and $\boldsymbol{b}$ satisfy the same linear recurrence, then $\boldsymbol{a} + \boldsymbol{b}$ also satisfies the same recurrence. ■

The property implied by Proposition 1 is called ***the shift-and-add property*** for sequences generated by the same polynomial.

Corollary 1: The sequences generated by a polynomial $f(x)$ are closed under bitwise addition.

Definition 6: The ***exponent*** $e(h(x))$ of a polynomial $h(x)$, where $h(0) \neq 0$, is the least integer e such that $h(x)$ divides $x^e - 1$.

Definition 7: An irreducible polynomial of degree n that generates a sequence of period $2^n - 1$, is called a ***primitive polynomial***. The sequence of period $2^n - 1$ which it generates is called a (***span*** n) **M-*sequence***.

Definition 8: A ***zero factor*** $\mathrm{ZF}(n, k)$ (with ***exponent*** k) is a set of d sequences of period k, which contains each nonzero n-tuple exactly once as a window in one of the sequences.

All known parameters n, d, and k of zero factors, can be inferred from the following proposition proved in [13].

Proposition 2: Let $f_i(x)$, $1 \leq i \leq r$, be r distinct irreducible polynomials of degree n and exponent e. Then the characteristic polynomial $\prod_{i=1}^{r} f_i(x)$ generates a zero factor with exponent e.

All zero factors whose sequences are generated by polynomials are presented in Proposition 2.

Definition 9: For a polynomial $f(x)$, let $S(f(x))$ denote the set of sequences whose characteristic polynomial is $f(x)$. The polynomial $f(x)$ ***generates*** these sequences and each sequence $\boldsymbol{a}$ in $S(f(x))$ is ***generated*** by $f(x)$.

Corollary 2: Let $f_i(x)$, $1 \leq i \leq r$, be r distinct irreducible polynomials of degree n and exponent e. The polynomial $\prod_{i=1}^{r} f_i(x)$ generates all the sequences which are generated by the polynomial $\prod_{i=1}^{r-1} f_i(x)$. Each $((r-1)n)$-tuples is contained as a window in exactly one sequence generated by $\prod_{i=1}^{r-1} f_i(x)$. Each (rn)-tuples is contained as a window in exactly one sequence generated by $\prod_{i=1}^{r} f_i(x)$.

Sometimes it will be important to distinguish between primitive polynomials and an irreducible polynomials which are not primitive and hence such an irreducible polynomial will be called an ***only-irreducible polynomial***.

III. Folding Sequences into Arrays

The folding technique which will be used in this paper to construct an $(r, t; n, m)$-PRAC (and also an $(r, t; n, m)$-PRA) is based on folding of nonzero sequences generated by polynomials which generate zero factors with exponent $r \cdot t$ into $r \times t$ arrays. The construction was first presented by MacWilliams and Sloane [16] to construct a PRA from an M-sequence. Assume that $\eta = 2^{nm} - 1$, $r = 2^n - 1$, and

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 14:54:01 UTC from IEEE Xplore. Restrictions apply.

$t = \frac{n}{r}$, where $\gcd(r,t) = 1$. Let $\mathcal{S} = s_0 s_1 s_2 \cdots$ be a span nm M-sequence. Write the consecutive elements of $\mathcal{S}$ down the right diagonals of an $r \times t$ array $\mathcal{B} = \{b_{ij}\}$, $0 \le i \le r-1$, $0 \le j \le t-1$, starting at b_{00}, b_{11}, b_{22} and so on, where the last position is $b_{r-1,t-1}$. After writing position b_{ij} we continue to write $b_{i+1,j+1}$, where $i+1$ is taken modulo r and $j+1$ is taken modulo t. In other words, s_k will be written in position b_{ij}, where $i \equiv k \pmod r$ and $j \equiv k \pmod t$. By the well-known Chinese remainder theorem [7, p. 11] each k, where $0 \le k \le rt-1$, has a different solution for these two equations. This method is called ***folding***. Another type of folding for pseudo-random arrays was presented in [26]. A general technique of folding into various shapes was considered in [6]. The following proposition was proved in [16].

Proposition 3: Each $n \times m$ nonzero matrix is contained exactly once as a window in the $r \times t$ array $\mathcal{B}$ obtained by folding a span nm M-sequence, where $r = 2^n - 1$ and $t = (2^{nm}-1)/r$. Moreover, $\mathcal{B}$ has the shift-and-add property, that is, $\mathcal{B}$ is an $(r, t; n, m)$-PRA.

If the length of a sequence $\mathcal{S}$ is rt, where $\gcd(r,t) = 1$ we denote by $\mathcal{F}(\mathcal{S}; r, t)$ the matrix $\mathcal{B}$ of size $r \times t$ obtained by folding $\mathcal{S}$ into $\mathcal{B}$.

Example 1: For $n = m = 2$, $r = 3$, and $t = 5$, consider the span 4 M-sequence $\mathcal{S} = [000111101011001]$, with positions numbered from $0, 1$, up to 14. Consider now the 3×5 array $\mathcal{B}$ with the entries b_{ij}, $0 \le i \le 2$, $0 \le j \le 4$, where the positions 0 through 14, of the sequence, are folded into $\mathcal{B}$ as follows

$$\begin{bmatrix} b_{00} & b_{01} & b_{02} & b_{03} & b_{04} \\ b_{10} & b_{11} & b_{12} & b_{13} & b_{14} \\ b_{20} & b_{21} & b_{22} & b_{23} & b_{24} \end{bmatrix}, \quad \begin{bmatrix} 0 & 6 & 12 & 3 & 9 \\ 10 & 1 & 7 & 13 & 4 \\ 5 & 11 & 2 & 8 & 14 \end{bmatrix}.$$

The M-sequence $\mathcal{S}$ is folded into the array $\mathcal{B}$ keeping the order of the entries in $\mathcal{S}$ according to the order defined by $\mathcal{B}$, i.e.,

$$\mathcal{F}(\mathcal{S}; 3, 5) = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

which forms a $(3, 5; 2, 2)$-PRA. ■

If the array $\mathcal{B}$ constructed in this way by folding a sequence $\mathcal{S}$ will be shifted horizontally or/and vertically, it will be equivalent to an array obtained by folding $\mathcal{S}$ into the array from another point of $\mathcal{S}$. Since the sequences obtained from any primitive polynomial have the shift-and-add property, it follows that if we add the array with its folded sequence to one of its nontrivial shifts, the outcome will be another nontrivial shift of the array $\mathcal{B}$. This property is the shift-and-add property of the array. This property is also derived by observing that two arrays obtained by folding two sequences preserve the bitwise addition of the sequences. Similar property is satisfied for the bitwise product. This is summarized in the following lemma.

Lemma 2: If $\boldsymbol{u}$ and $\boldsymbol{v}$ are sequences of length $r \cdot t$ such that $\gcd(r,t) = 1$, then

1) $\mathcal{F}(\boldsymbol{u}; r, t) + \mathcal{F}(\boldsymbol{v}; r, t) = \mathcal{F}(\boldsymbol{u} + \boldsymbol{v}; r, t)$;
2) $\mathcal{F}(\boldsymbol{u}; r, t) \cdot \mathcal{F}(\boldsymbol{v}; r, t) = \mathcal{F}(\boldsymbol{u} \cdot \boldsymbol{v}; r, t)$.

Example 2: Consider the array and the M-sequence $\mathcal{S}$ of Example 1. We shift the array horizontally by 2 and vertically by 1 and add them as follows; the first bit of $\mathcal{S}$ is in bold.

$$\begin{bmatrix} \mathbf{0}\,1\,0\,1\,0 \\ 1\,0\,0\,0\,1 \\ 1\,1\,0\,1\,1 \end{bmatrix} + \begin{bmatrix} 1\,1\,1\,1\,0 \\ 1\,0\,\mathbf{0}\,1\,0 \\ 0\,1\,1\,0\,0 \end{bmatrix} = \begin{bmatrix} 1\,0\,1\,0\,\mathbf{0} \\ 0\,0\,0\,1\,1 \\ 1\,0\,1\,1\,1 \end{bmatrix}.$$

The M-sequence $\mathcal{S}$ starts in the leftmost array in b_{00}, in the middle array at b_{12}, and in the rightmost array at b_{04}. ■

Proposition 3 proved in [16] is generalized to obtain more PRAs in the following theorem, which is our first main result.

Theorem 1: Let n, m, r, and t, be positive integers such that $rt = 2^{nm} - 1$ and $\gcd(r,t) = 1$. If r divides $2^n - 1$ and the n integers, 2^i, $0 \le i \le n-1$, are distinct modulo r, then the folding of sequences generated by a primitive polynomial whose length is rt into an $r \times t$ arrays form an $(r, t; n, m)$-PRA.

IV. Folding Sequences of a Zero Factor

Let $\mathcal{C}$ be a set of sequences of length rt, such that $\gcd(r,t) = 1$. We denote by $\mathcal{F}(\mathcal{C}; r, t)$ the set of $r \times t$ arrays obtained by folding the sequences in $\mathcal{C}$ into $r \times t$ arrays.

Example 3: Let $r = 3$, $t = 7$, $n = 2$, and $m = 3$. Let $f(x) = x^6 + x^5 + x^4 + x^2 + 1$ be the irreducible polynomial of degree 6 and exponent 21, i.e., it generates a ZF$(6, 21)$ $\mathcal{C}$. The nonzero sequences of $\mathcal{C}$ generated by $f(x)$ are $[00000101000100110010 11]$, $[01000011111011010101 11]$, and $[00100011011111100111 0]$. Folding these sequences of $\mathcal{C}$ into 3×7 arrays yields the following three arrays of $\mathcal{F}(\mathcal{C}; 3, 7)$:

$$\left\{ \begin{bmatrix} 0\,0\,0\,0\,0\,0\,0 \\ 1\,0\,0\,1\,0\,1\,1 \\ 1\,0\,0\,1\,0\,1\,1 \end{bmatrix}, \begin{bmatrix} 0\,0\,1\,0\,1\,1\,1 \\ 1\,1\,1\,0\,0\,1\,0 \\ 1\,1\,0\,0\,1\,0\,1 \end{bmatrix}, \begin{bmatrix} 0\,0\,1\,0\,1\,1\,1 \\ 1\,0\,0\,1\,0\,1\,1 \\ 1\,0\,1\,1\,1\,0\,0 \end{bmatrix} \right\}.$$

These three arrays form a $(3, 7; 2, 3)$-PRAC. ■

Each row of the arrays in Example 3 is either the M-sequence $[0010111]$ or the all-zero sequence. Each column is either the M-sequence $[011]$ or the all-zero sequence. The following lemma used to prove this property is easily verified.

Lemma 3: Let r and t be two positive integers, where $\gcd(r,t) = 1$ and let $\mathcal{S} = (\boldsymbol{a} \cdots \boldsymbol{a})$ be a sequence of length rt.

1) If the length of $\boldsymbol{a}$ is t, then each row of $\mathcal{F}(\mathcal{S}; r, t)$ is given by $\boldsymbol{a}$ and each column is either the all-one sequence or the all-zero sequence.
2) If the length of $\boldsymbol{a}$ is r, then each column of $\mathcal{F}(\mathcal{S}; r, t)$ is given by $\boldsymbol{a}$ and each row is either the all-one sequence or the all-zero sequence.

Example 4: If $\mathcal{S} = [01101101101101101101 1]$, then

$$\mathcal{A} \triangleq \mathcal{F}(\mathcal{S}; 3, 7) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$ ■

Example 5: If $\mathcal{S} = [10010111001011100101 1]$, then

$$\mathcal{B} \triangleq \mathcal{F}(\mathcal{S}; 3, 7) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}.$$ ■

Lemma 4: Let $\mathcal{A}$ be an $r \times t$ array for which each column is given by a sequence $\boldsymbol{u}$ and each row is either the all-one sequence or the all-zero sequence. Let $\mathcal{B}$ be an $r \times t$ array for which each row is given by a sequence $\boldsymbol{v}$ and each column is either the all-one sequence or the all-zero sequence. Then, the bitwise product $\mathcal{A} \cdot \mathcal{B}$ is an $r \times t$ array for which each row is either an all-zero sequence or the sequence $\boldsymbol{v}$ and each column of $\mathcal{A} \cdot \mathcal{B}$ is either an all-zero sequence or the sequence $\boldsymbol{u}$.

Example 6: Let $\mathcal{A}$ and $\mathcal{B}$ be the 3×7 arrays of Example 4 and 5, respectively. The bitwise product of $\mathcal{A}$ and $\mathcal{B}$ is

$$\mathcal{A} \cdot \mathcal{B} = \begin{bmatrix} 0\,0\,0\,0\,0\,0\,0 \\ 1\,1\,1\,1\,1\,1\,1 \\ 1\,1\,1\,1\,1\,1\,1 \end{bmatrix} \cdot \begin{bmatrix} 1\,0\,0\,1\,0\,1\,1 \\ 1\,0\,0\,1\,0\,1\,1 \\ 1\,0\,0\,1\,0\,1\,1 \end{bmatrix} = \begin{bmatrix} 0\,0\,0\,0\,0\,0\,0 \\ 1\,0\,0\,1\,0\,1\,1 \\ 1\,0\,0\,1\,0\,1\,1 \end{bmatrix}.$$

■

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 14:54:01 UTC from IEEE Xplore. Restrictions apply.

For the rest of this section, let $f_i(x)$, $i = 1, 2$, be a polynomial of degree n_i and exponent r_i.

Lemma 5: If $\mathcal{A}$ and $\mathcal{B}$ are two distinct $r_1 \times r_2$ arrays whose columns are either all-zero sequences or a sequences of length r_1 contained in $S(f_1(x))$ and whose rows are either all-zero sequences or sequences of length r_2 contained in $S(f_2(x))$, then the addition $\mathcal{A}+\mathcal{B}$ is also an $r_1 \times r_2$ array whose columns are either all-zero sequences or sequences contained in $S(f_1(x))$ and whose rows are either all-zero sequences or a sequences contained in $S(f_2(x))$.

Example 7: Consider the primitive polynomial $f_1(x) = x^2 + x + 1$ whose zero factor has exactly one sequence [011] and the reducible polynomial $f_2(x) = x^6+x^5+x^4+x^3+x^2+x+1 = (x^3+x^2+1)(x^3+x+1)$ whose zero factor $\mathcal{F}$ has nine sequences [0011101], [0010111], [0000101], [0101011], [0001111], [0111111], [0001001], [0000011], [0011011]. In the following 3×7 array $\mathcal{A}$, each column is either all-zero sequence or the sequence [011] (or a shift of it). Each row is either the all-zero sequence or the sequence [0001111] (or a shift of it).

$$\mathcal{A} \triangleq \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Adding $\mathcal{A}$ to one of its shifts $\mathcal{B}$ we obtain the array $\mathcal{T}_1$:

$$\mathcal{A}+\mathcal{B} = \begin{bmatrix} 0\,0\,0\,0\,0\,0\,0 \\ 0\,0\,0\,1\,1\,1\,1 \\ 0\,0\,0\,1\,1\,1\,1 \end{bmatrix} + \begin{bmatrix} 0\,0\,1\,1\,1\,1\,0 \\ 0\,0\,1\,1\,1\,1\,0 \\ 0\,0\,0\,0\,0\,0\,0 \end{bmatrix} = \begin{bmatrix} 0\,0\,1\,1\,1\,1\,0 \\ 0\,0\,1\,0\,0\,0\,1 \\ 0\,0\,0\,1\,1\,1\,1 \end{bmatrix}.$$

Adding $\mathcal{T}_1$ to one of its shifts $\mathcal{T}_2$ we obtain the array:

$$\mathcal{T}_1+\mathcal{T}_2 = \begin{bmatrix} 0\,0\,1\,1\,1\,1\,0 \\ 0\,0\,1\,0\,0\,0\,1 \\ 0\,0\,0\,1\,1\,1\,1 \end{bmatrix} + \begin{bmatrix} 0\,1\,1\,1\,1\,0\,0 \\ 1\,1\,1\,1\,0\,0\,0 \\ 1\,0\,0\,0\,1\,0\,0 \end{bmatrix} = \begin{bmatrix} 0\,1\,0\,0\,0\,1\,0 \\ 1\,1\,0\,1\,0\,0\,1 \\ 1\,0\,0\,1\,0\,1\,1 \end{bmatrix}.$$

Each column of $\mathcal{A}+\mathcal{B}$ and of $\mathcal{T}_1+\mathcal{T}_2$ is either the all-zero sequence or the sequence [011] and each row is either the all-zero sequence or a sequence from $\mathcal{F}$. ■

Let $S(f_1(x))$ and $S(f_2(x))$ be the set of nonzero sequences generated by $f_1(x)$ and $f_2(x)$, respectively, and all the sequences in $S(f_i(x))$ have period r_i, $i = 1, 2$ and $\gcd(r_1, r_2) = 1$. Let

$$\mathbb{C} \triangleq \{\mathcal{F}(\boldsymbol{a}\cdot\boldsymbol{b}; r_1, r_2) \ : \ \boldsymbol{a} \in S(f_1(x)), \quad \boldsymbol{b} \in S(f_2(x))\}\ .$$

Lemma 6: Let $\boldsymbol{a}$ be a nonzero sequence generated by $f_1(x)$ and let $\boldsymbol{b}$ be a nonzero sequence generated by $f_2(x)$. If $\mathcal{S}_1 \triangleq ([\boldsymbol{a}\,\boldsymbol{a}\,\cdots\,\boldsymbol{a}] \in \mathbb{F}_2^{r_1\cdot r_2}$ and $\mathcal{S}_2 \triangleq ([\boldsymbol{b}\,\boldsymbol{b}\,\cdots\,\boldsymbol{b}] \in \mathbb{F}_2^{r_1\cdot r_2}$, then the $r_1 \times r_2$ array $\mathcal{F}(\mathcal{S}_1; r_1, r_2)\cdot\mathcal{F}(\mathcal{S}_2; r_1, r_2)$ is contained in $\mathcal{C}$.

Corollary 3: If $\tilde{\boldsymbol{a}}$ is an n_1-tuple in the sequence $\boldsymbol{a}$ and $\tilde{\boldsymbol{b}}$ is an n_2-tuple in the sequence $\boldsymbol{b}$, then the $n_1 \times n_2$ matrix M whose i-th row is $\tilde{\boldsymbol{b}}$ if the i-th entry of $\tilde{\boldsymbol{a}}$ is *one* and all-zero if the i-th entry of $\tilde{\boldsymbol{a}}$ is *zero*, is contained in $\mathcal{F}(\boldsymbol{a}\cdot\boldsymbol{b}; r_1, r_2)$.

Corollary 4: If $\tilde{\boldsymbol{a}}$ is an n_1-tuple in the sequence $\boldsymbol{a}$ and $\tilde{\boldsymbol{b}}$ is an n_2-tuple in the sequence $\boldsymbol{b}$, then the $n_1 \times n_2$ matrix M whose i-th column is $\tilde{\boldsymbol{a}}$ if the i-th entry of $\tilde{\boldsymbol{b}}$ is *one* and all-zero if the i-th entry of $\tilde{\boldsymbol{b}}$ is *zero*, is contained in $\mathcal{F}(\boldsymbol{a}\cdot\boldsymbol{b}; r_1, r_2)$.

Lemma 7: Let $S(f_1(x))$ and $S(f_1(x))$ be the set of nonzero sequences generated by $f_1(x)$ and $f_2(x)$, respectively. If the sequences in $S(f_i(x))$ have period r_i, $i = 1, 2$ and $\gcd(r_1, r_2) = 1$, then $r_1 r_2$ is the period of the sequences in the linear span of

$$\mathcal{C} \triangleq \{\boldsymbol{a}\cdot\boldsymbol{b} \ : \ \boldsymbol{a} \in S(f_1(x)), \quad \boldsymbol{b} \in S(f_2(x))\}\ .$$

Definition 10: For two polynomials $f_1(x)$ and $f_2(x)$ we denote by $f_1(x) \vee f_2(x)$ the polynomial whose roots have the form $\alpha\beta$, where α is a root of $f_1(x)$ and β is a root of $f_2(x)$ in the splitting field of $f_1(x)f_2(x)$.

Example 8: Let $f_1(x) = x^4+x+1$ and $f_2(x) = x^3+x+1$. The splitting field of $f_1(x)f_2(x)$ is the finite field $\mathbb{F}_{2^{12}}$. Let α be a primitive element of $\mathbb{F}_{2^{12}}$ with the minimal ploynomial $m(x) = x^{12}+x^7+x^6+x^5+x^3+x+1$. The roots of $f_1(x)$ are $\beta_1 = \alpha^{273}, \beta_2 = \alpha^{546}, \beta_3 = \alpha^{1092}, \beta_4 = \alpha^{2184}$. The roots of $f_2(x)$ are $\gamma_1 = \alpha^{585}, \gamma_2 = \alpha^{1170}, \gamma_3 = \alpha^{2340}$. Finally, $f_1(x) \vee f_2(x) = \prod_{i=1}^{4}\prod_{j=1}^{3}(x-\beta_i\gamma_j) = x^{12}+x^9+x^5+x^4+x^3+x+1$. ■

Definition 11: Let $S(f_1(x))\cdot S(f_2(x))$ be defined as a vector space spanned by all products $\boldsymbol{a}\cdot\boldsymbol{b}$, where $\boldsymbol{a} \in S(f_1(x))$ and $\boldsymbol{b} \in S(f_2(x))$.

The following proposition implied by Theorem 8.67 in [14] is a key result for the construction of PRACs.

Proposition 4: Let $f_i(x)$, $i = 1, 2$, be a polynomial of degree n_i generating a ZF(n_i, r_i). Let β_i, $1 \le i \le m$ be the roots of $f_1(x)$ and γ_j, $1 \le j \le n$ be the roots of $f_2(x)$ in the splitting field of $f_1(x)f_2(x)$. Then $S(f_1(x))\cdot S(f_2(x)) = S(f_1(x) \vee f_2(x))$, where

$$f_1(x) \vee f_2(x) = \prod_{i=1}^{m}\prod_{j=1}^{n}(x-\beta_i\gamma_j).$$

Proposition 4 implies in other words that the set of sequences generated by $f_1(x) \vee f_2(x)$ is the linear span of the sequences that are obtained as a bitwise product of the sequences generated by $f_1(x)$ and $f_2(x)$. Proposition 4 leads to our second main theorem and a second construction of PRACs.

Theorem 2: Let r_1, r_2, n_1, n_2 be positive integers such that $\gcd(r_1, r_2) = 1$, and let $f_i(x)$, $i = 1, 2$, be a polynomial of degree n_i and exponent r_i. If the characteristic polynomial $g(x) \triangleq f_1(x) \vee f_2(x)$ generates a zero factor ZF$(n_1 n_2, r_1 r_2)$, then folding the sequences generated by $g(x)$ into $r_1 \times r_2$ arrays yields an $(r_1, r_2; n_1, n_2)$-PRAC with $(2^{n_1 n_2}-1)/r_1 r_2$ codewords.

Lemma 7 and Theorem 2 imply the following consequence.

Corollary 5: Let r_i, n_i, $i = 1, 2$, be positive integers such that $\gcd(r_1, r_2) = 1$. Let $f_i(x)$ be a polynomial of degree n_i and exponent r_i. Then, the sequences generated by $g(x) \triangleq f_1(x) \vee f_2(x)$ can be folded into $r_1 \times r_2$ arrays and form an $(r_1, r_2; n_1, n_2)$-PRAC.

Example 9: Let $f_1(x) = (x^3+x^2+1)(x^3+x+1) = x^6+x^5+x^4+x^3+x^2+x+1$ be a reducible polynomial with exponent 7 and $f_2(x) = (x^4+x^3+1)(x^4+x+1) = x^8+x^7+x^5+x^4+x^3+x+1$ be a reducible polynomial with exponent 15. The polynomial $f_1(x) \vee f_2(x) = (x^{12}+x^8+x^6+x^5+x^3+x^2+1)(x^{12}+x^9+x^5+x^4+x^3+x+1)(x^{12}+x^{10}+x^9+x^7+x^6+x^4+1)(x^{12}+x^{11}+x^9+x^8+x^7+x^3+1)$ is a reducible polynomial with exponent 105. By Corollary 5, folding the sequences of this zero factor into 7×15 arrays yields a $(7, 15; 6, 8)$-PRAC. ■

Another generalization of Theorem 1 and a comparison between the various constructions is presented in [4].

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 14:54:01 UTC from IEEE Xplore. Restrictions apply.

V. Hierarchies of Codes

Theorem 2 and its consequence Corollary 5, imply a variety of PRACs. In this section, we examine which combinations for $f_1(x)$, $f_2(x)$, and $g(x)$ are possible, and which cannot occur. The folded sequences are those generated by $g(x)$.

For this purpose, we will have to use the following theorem proved in [14, p. 85, Theorem 3.5].

Theorem 3: The degree of irreducible polynomial with exponent e is the multiplicative order of 2 modulo e.

Corollary 6: All the irreducible polynomials with exponent e have the same degree n. All reducible polynomials with exponent e are product of irreducible polynomials of degree n and exponent e.

Theorem 3 and Corollary 6 enable us to decide which combinations of the three type of polynomials for $f_1(x)$, $f_2(x)$ and $g(x)$ are not possible.

Lemma 8: The polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 2 and Corollary 5 cannot be a primitive polynomial.

Lemma 9: If $f_1(x)$ or $f_2(x)$ is reducible polynomial, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 2 is a reducible polynomial.

Example 10: Let $f_1(x) = \sum_{i=1}^{6} x^i + 1 = (x^3 + x + 1)(x^3 + x^2 + 1)$ be a reducible polynomial with exponent 7 and $f_2(x) = \sum_{i=1}^{10} x^i + 1$ be an only-irreducible polynomial with exponent 11.

$f_1(x) \vee f_2(x) = (x^{30} + x^{28} + x^{27} + x^{26} + x^{23} + x^{21} + x^{20} + x^{19} + x^{16} + x^{14} + x^{13} + x^{12} + x^9 + x^8 + x^7 + x^4 + x^2 + x + 1)(x^{30} + x^{29} + x^{28} + x^{26} + x^{23} + x^{22} + x^{21} + x^{18} + x^{17} + x^{16} + x^{14} + x^{11} + x^{10} + x^9 + x^7 + x^4 + x^3 + x^2 + 1)$. These two factors of $f_1(x) \vee f_2(x)$ are only-irreducible polynomials with exponent $77 = 7 \cdot 11$ and therefore $f_1(x) \vee f_2(x)$ forms a ZF$(60, 77)$. By Theorem 2 and Corollary 5, the folding of its sequences into 7×11 arrays is a $(7, 11; 6, 10)$-PRAC. ■

Example 11: Let $f_1(x) = \sum_{i=0}^{6} x^i = (x^3+x^2+1)(x^3+x+1)$ be a reducible polynomial with exponent 7 and let $f_2(x) = x^2 + x + 1$ be a primitive polynomial with exponent 3. The polynomial $f_1(x) \vee f_2(x) = (x^6 + x^4 + x^2 + x + 1)(x^6 + x^5 + x^4 + x^2 + 1)$ is a reducible polynomial with exponent 7. By Corollary 5 folding its sequences into 7×3 arrays is a $(7, 3; 6, 2)$-PRAC. ■

Lemma 10: If $f_i(x)$, $i = 1, 2$, is a polynomial of degree n_i and exponent r_i, such that $\gcd(n_1, n_2) > 1$, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 2 is a reducible polynomial.

Corollary 7: If $f_i(x)$, $i = 1, 2$ are polynomials of degree n_i with exponent r_i, such that $\gcd(n_1, n_2) > 1$ and $f_1(x)$ or $f_2(x)$ is an only-irreducible polynomial, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Corollary 5 is a reducible polynomial.

Example 12: Let $f_1(x) = x^4 + x^3 + x^2 + x + 1$ be an only-irreducible polynomial with exponent 5 and let $f_2(x) = x^6 + x^3 + 1$ be an only-irreducible polynomial with exponent 9. The polynomial $f_1(x) \vee f_2(x) = x^{24} + x^{21} + x^{15} + x^{12} + x^9 + x^3 + 1 = (x^{12} + x^9 + 1)(x^{12} + x^3 + 1)$ is a reducible polynomial with exponent 45. By Corollary 5, folding its sequences into 5×9 arrays is a $(5, 9; 4, 6)$-PRAC. ■

Example 13: Let $f_1(x) = x^4 + x^3 + x^2 + x + 1$ be an only-irreducible polynomial with exponent 5 and $f_2(x) = x^6 + x^5 + 1$ be a primitive polynomial with exponent 63.

$f_1(x) \vee f_2(x) = (x^{12} + x^4 + x^2 + x + 1)(x^{12} + x^{11} + x^{10} + x^9 + x^8 + x^6 + x^3 + x + 1)$. These two factors of $f_1(x) \vee f_2(x)$ are irreducible polynomials with exponent $315 = 5 \cdot 63$ and therefore $f_1(x) \vee f_2(x)$ form a ZF$(24, 315)$. By Corollary 5, the folding of its sequences into 5×63 arrays is a $(5, 63; 4, 6)$-PRAC. ■

Corollary 8: If $f_i(x)$, $i = 1, 2$, is an irreducible polynomial of degree n_i and exponent r_i such that $\gcd(n_1, n_2) = 1$, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Corollary 5 is an only-irreducible polynomial.

Example 14: Let $f_1(x) = x^4 + x^3 + x^2 + x + 1$ be an only-irreducible polynomial with exponent 5 and $f_2(x) = x^9 + x + 1$ be an only-irreducible polynomial with exponent 73.

$f_1(x) \vee f_2(x) = x^{36} + x^{28} + x^{27} + x^{20} + x^{18} + x^{12} + x^{10} + x^9 + x^4 + x^3 + x^2 + x + 1$ is an only-irreducible polynomial with exponent $365 = 5 \cdot 73$. By Corollary 5, folding its sequences into 5×73 arrays is a $(5, 73; 4, 9)$-PRAC. ■

Example 15: Let $f_1(x) = \sum_{i=1}^{4} x^i$ be an only-irreducible polynomial with exponent 5 and Let $f_2(x) = x^3 + x^2 + 1$ be a primitive polynomial with exponent 7. The polynomial $f_1(x) \vee f_2(x) = x^{12} + x^{11} + x^{10} + x^8 + x^5 + x^4 + x^3 + x^2 + 1$ is an only-irreducible polynomial with exponent $35 = 7 \cdot 5$. By Corollary 5, folding its sequences into 5×7 arrays is a $(5, 7; 4, 3)$-PRAC. ■

Example 16: Let $f_1(x) = x^4 + x + 1$ and $f_2(x) = x^3 + x + 1$ be two primitive polynomials with exponents 15 and 7, respectively. $f_1(x) \vee f_2(x) = x^{12} + x^9 + x^5 + x^4 + x^3 + x + 1$ (See Example 8) is an only-irreducible polynomial with exponent $105 = 15 \cdot 7$. By Corollary 5, folding the sequences of this zero factor into 15×7 arrays is an $(15, 7; 4, 3)$-PRAC. ■

There are no more possible combinations of primitive, irreducible, and reducible polynomials for $f_1(x)$, $f_2(x)$, and $g(x)$, beside those which are summarized in Table I.

A second hierarchy of PRACs can be inferred from the following lemma.

Lemma 11: Let $f_1(x)$ be an irreducible polynomial of degree n and exponent r_1 and let $h_i(x)$, $1 \leq i \leq \ell$, $\ell \geq 2$ be ℓ irreducible polynomials of degree m and exponent r_2. The arrays of the $(r_1, r_2; n, \ell m)$-PRAC generated by folding the sequences generated by $f_1(x) \vee \prod_{i=1}^{\ell} h_i(x)$ into $r_1 \times r_2$ arrays contain the arrays of the $(r_1, r_2; n, (\ell - 1)m)$-PRAC generated by folding the sequences generated by $f_1(x) \vee \prod_{i=1}^{\ell-1} h_i(x)$ into $r_1 \times r_2$ arrays.

Finally, Simon R. Blackburn has extended the results for folding of reducible polynomials.

TABLE I
Types of PRACs

$f_1(x)$	$f_2(x)$	$g(x)$	references
reducible	reducible	reducible	Lemma 9, Example 9
reducible	only-irreducible	reducible	Lemma 9, Example 10
reducible	primitive	reducible	Lemma 9, Example 11
only-irreducible	only-irreducible	reducible	Corollary 7, Example 12
only-irreducible	only-irreducible	only-irreducible	Corollary 8, Example 14
only-irreducible	primitive	reducible	Corollary 7, Example 13
only-irreducible	primitive	only-irreducible	Corollary 8, Example 15
primitive	primitive	only-irreducible	Corollary 8, Example 16

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 14:54:01 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] M. BLAUM AND J. BRUCK, *MDS array codes for correcting criss-cross errors*, *IEEE Trans. on Infor. Theory,* 46 (2000) 1068–1077.

[2] M. BLAUM, J. BRUCK, A. VARDY, *Interleaving schemes for multidimensional cluster errors*, *IEEE Trans. on Infor. Theory,* 44 (1998) 730–743.

[3] M. BREITBACH, M. BOSSERT, V. ZYABLOV, V. SIDORENKO, *Array codes correcting a two-dimensional cluster of errors*, *IEEE Trans. on Infor. Theory,* 44 (1998) 2025–2031.

[4] Y. M. CHEE, T. ETZION, AND H. LAO, *On de Bruijn array codes, part II: Linear codes*, *arxiv.org/abs/2501.12124*, February 2025.

[5] T. ETZION, *Constructions for perfect maps and pseudo-random arrays*, *IEEE Trans. on Infor. Theory,* 34 (1988) 1308–1316.

[6] T. ETZION, *Sequence folding, lattice tiling, and multidimensional coding*, *IEEE Trans. on Infor. Theory,* 57 (2011) 4383–4400.

[7] T. ETZION, *Sequences and the de Bruijn Graph: Properties, Constructions, and Applications*, *London, UK; San Diego, US, Cambridge, US: Elsevier, 2024*.

[8] T. ETZION, *Pseudo-Random and de Bruijn Array Codes*, *IEEE Int. Symp. on Inf. Theory (ISIT)*, pp. 1742–1747, Athens, Greece, July 2024.

[9] T. ETZION, *On de Bruijn array codes, Part I: nonlinear codes*, *IEEE Trans. on Infor. Theory,* 71 (2025) 1434–1449.

[10] T. ETZION AND A. VARDY, *Two-dimensional interleaving schemes with repetitions: Constructions and bounds*, *IEEE Trans. on Infor. Theory,* 48 (2002) 428–457.

[11] T. ETZION AND E. YAAKOBI, *Error-correction of multidimensional bursts*, *IEEE Trans. on Infor. Theory,* 55 (2009) 961–976.

[12] J. GENG, *Structured-light 3D surface imaging: A tutorial*, *Adv. Opt. Photon.,* 3 (2011) 128–160.

[13] S. W. GOLOMB, *Shift Register Sequences,* San Francisco, CA: Holden Day, 1967; 2nd revised edition, Laguna Hills, CA: Aegean Park, 1980; 3rd revised edition, World Scientific, Singapore 2017.

[14] R. LIDL, AND H. NIEDERREITER, *Finite fields*, *Cambridge, UK: Cambridge Univ. Press, 1997*.

[15] H. LIN, L. NIE, AND Z. SONG, *A single-shot structured light means by encoding both color and geometrical features*, *Pattern Recognition,* 54 (2016) 178–189.

[16] F. J. MACWILLIAMS AND N. J. A. SLOANE, *Pseudo-random sequences and arrays*, *Proceedings of the IEEE,* 64 (1976) 1715–1729.

[17] C. J. MITCHELL, *Aperiodic and semi-periodic perfect maps*, *IEEE Trans. on Infor. Theory,* 41 (1995) 88–95.

[18] T. NOMURA, H. MIYAKAWA, H. IMAI, AND A. FUKUDA, *The theory of two-dimensional linear recurring arrays*, *IEEE Trans. on Infor. Theory,* 18 (1972) 773–785.

[19] R. A. MORANO, C. OZTURK, R. CONN, S. DUBIN, S. ZIETZ, AND J. NISSANOV, *Structured light using pseudorandom codes*, *IEEE Trans. on Pattern Analysis and Machine Intelligence,* 20 (1998) 322–327.

[20] K. G. PATERSON, *Perfect maps*, *IEEE Trans. on Infor. Theory,* 40 (1994) 743–753.

[21] I. S. REED AND R. M. STEWART, *Note on existence of perfect maps*, *IRE Trans. on Infor. Theory,* 8 (1962) 10–12.

[22] R. M. ROTH, *Maximum-rank array codes and their application to crisscross error correction*, *IEEE Trans. on Infor. Theory,* 37 (1991) 328–336.

[23] J. SALVI, S. FERNANDEZ, T. PRIBANIC, AND X. LLADO, *A state of art in structured light patterns for surface profilometry*, *Pattern Recognition,* 43 (2010) 2666–2680.

[24] J. SALVI, J. PAGÈS, AND J. BATLLE, PATTERN CODIFICATION STRATEGIES IN STRUCTURED LIGHT SYSTEMS, PATTERN RECOGNITION, 37 (2004) 827–849.

[25] M. SCHWARTZ AND J. BRUCK, *Constrained codes as networks of relations*, *IEEE Trans. on Infor. Theory,* 54 (2008) 2179–2195.

[26] R. SPANN, *A two-dimensional correlation property of pseudo-random maximum-length sequences*, *Proceedings of the IEEE,* 53 (1963) 2137.

[27] I. TAL, T. ETZION, AND R. ROTH, *On row-by-row coding for 2-D constraints*, *IEEE Trans. on Infor. Theory,* 55 (2009) 3565–3576.

Authorized licensed use limited to: Lakehead University. Downloaded on September 24,2026 at 14:54:01 UTC from IEEE Xplore. Restrictions apply.